Functional Specifications & Business Logic Reference

This document provides a detailed listing and explanation of all core functions, workflows, and automated logic within the Techser Plant Management ERP.

1. Inventory & Warehouse Management

A. Quick GRN (Goods Receipt Note)

- **Purpose:** To digitise the incoming material data post-unloading for quality control routing and financial logging.
- **Physical Process (Manual):** The physical operation of unloading all material from the truck and matching the items against the physical invoice copy provided by the supplier. This step is conducted manually before system entry.
- **Application Logic:** After the physical verification is complete, the user enters all items listed on the invoice into the application, including their specific quantities, make, and unit prices as provided on the invoice.
- **Inwarding Types:**
    - **Regular:** Standard purchase against an open Purchase Order.
    - **Replacement:** Goods sent by the vendor to replace previously rejected items. **Must be linked to a specific RTV Reference ID.**
- **Ledger Integration:**
    - **Regular Inward:** Automatically generates a "Pending Liability" entry in the Ledger.
    - **Replacement Inward:** Logs the arrival but does not create a double liability; it marks the original "Debit Note" as "Replacement in Progress."
- **Personnel Tracking & Audit:** Records **Received By**, **Entered By**, and a high-precision **Entry Timestamp**.
- **Data Flow:** This data is automatically routed to the **QC Management** module for technical inspection and audit.
- **Stock Status:** Items are placed in a 'Buffer Stock' (Hold Area). They are not available for production builds at this stage.

B. QC Management & RTV (Return to Vendor)

- **Purpose:** To ensure only high-quality components enter the assembly line and to reconcile financial liability.
- **Logic:** The Quality team audits the items currently in 'Buffer Stock.'
- **Personnel Tracking & Audit:** For every QC session, the system records **Audited By** and the exact **Audit Timestamp**.
- **Outcomes:**
    - **Accepted:** Items move to the Indexing phase.
    - **Rejected (RTV):**
        - **Rejected Registry:** All items that fail QC are listed in a dedicated **"RTV Dashboard"** for tracking and audit. This includes reference details such as the **Purchase Order (PO) Number**, **Supplier Invoice Number**, and **Original Indent Number**.
        - **Replacement Workflow:** The system provides an option to select these rejected items and send an official replacement request.

        - **Notification:** An automated email is sent to the supplier stating: *"The items listed below did not pass our Quality Control (QC) inspection and require immediate replacement."*
- **Ledger Update:**
    - Upon completion of the QC, the system automatically updates the initial pending entry in the **Ledger Book**.
    - If items are rejected, the system generates a **Debit Note** or reduces the pending liability amount to match only the **Accepted Quantity**.

C. Replacement Handling (The Audit Loop)

- **Purpose:** To handle the lifecycle of items returned by the vendor as replacements.
- **Inwarding Step:** Replacement items are entered via "Step A" (marked as Type: Replacement).
- **QC Enforcement:** The system triggers a **Mandatory QC Check** for replacements. They cannot bypass this to go directly to stock.
- **Ledger Finalisation:**
    - If **Replacement Passes QC:** The system automatically reverses the previous Debit Note or reinstates the liability for those units.
    - If **Replacement Fails QC:** The item is re-added to the RTV Dashboard, a new notification is sent, and the Debit Note remains active.
- **Activity Tracking:** Every step (Arrival -> QC -> Ledger Update) is logged with the user ID and a unique timestamp for that specific activity.

D. Hex Indexing & Unit Serialisation

- **Purpose:** To assign unique digital identities (Hex Codes or Serial Tags) and ensure absolute traceability without duplication.
- **Category 1: Serialised Tracking (1:1 Ratio)**
    - **High-Value Components & Spares:** Every single unit is assigned a unique **4-digit Hex Code**. One Hex Tag = One physical unit.
    - **Products:** Assigned a structured **Serial Number** (Rating Plate).
- **Category 2: Bulk/Standard Tracking (1:N Ratio)**
    - **Low-Value Components (e.g., Resistors, Capacitors, Screws):** These items **do not** receive individual Hex tags.
    - **Unique ID:** The **Item Code** (e.g., *R102505*) acts as the unique identifier.
    - **Quantity Logic:** A single Item Code can represent multiple units (e.g., 1000 nos). Stock is tracked as a cumulative balance for that Item Code + Make.
- **Technical Master Data:** For every item indexed (Spares, Components, or Products), the system maintains a detailed technical record containing. The system maintains specific technical records categorised by item type to ensure all relevant specifications are captured:
- **For Standard Components (Bulk):**
    - **Item Code:** Unique system identifier.
    - **Item Name:** Short description.
    - **Description:** Detailed technical details.
    - **Value:** Electrical/Physical value (e.g., 10K, 1/4W).
    - **Make:** Brand/Manufacturer.
    - **HSN Code:** Tax classification.
- **For Spares & High-Value Components (Serialized):**
    - **Hex Code:** Unique 4-digit ID.
    - **Item Name:** Specific spare part name.
    - **Ratings:** Capacity details (KVA).
    - **Volt:** DC Voltage specifications.

    - **Amp:** Current rating in Amperes.
    - **Revision:** Version/Design iteration.
    - **Add-On Part Number:** Supplementary internal reference.
    - **Specs Details:** Granular technical notes.
    - **Make & Price:** Brand and financial cost.
- **For Products (Serialised):**
    - **System Serial Number:** Unique product ID (Rating Plate).
    - **Product Name:** Standardised system name.
    - **Product Ratings:** Full system capacity.
    - **DC Bus:** Internal DC bus voltage details.
    - **Phase:** Input/Output phase configuration.
    - **Model & Series:** Catalogue classification.
    - **Specs Details:** Engineering specifications.
- **Common Tracking Fields (All Categories):**
    - **Status:** Current state (Buffer, Available, Consumed, Scrapped).
    - **Modifications & Timestamp:** Complete audit trail of record edits.

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- **Anti-Duplication Registry (The Master Guard):**
    - The system maintains a global registry of every generated code. Before printing any tag, the system performs a real-time check against the database to ensure the code does not already exist.
    - **Hex Tracking:** Every 4-digit code is recorded with its status (Generated, Printed, Indexed, Consumed, or Scrapped).
- **Tagging Logic:**
    - **Spares & High-Value Components:** Assigned a unique **4-digit Hexadecimal code** (0000 to FFFF). The system automatically picks the next available code in the sequence to prevent user error.
    - **Products:** Assigned a structured **Serial Number** (e.g., B26030001).
        - **Format:** `[Factory Code][Year][Month][4-Digit Sequence]`.
        - **Reset Logic:** The 4-digit sequence automatically resets to `0001` at the start of every new month for each factory.
- **Indexing Process:** Staff affix the physical tag and scan the QR (or type the 4-digit ID). The system "locks" this unique ID to the technical master data fields listed above and logs the **Indexing User** and **Timestamp**.
- **Stock Status:** Upon successful indexing, the status changes from 'Buffer Stock' to **'Available Stock'**.

2. Procurement & Supplier Logic

A. Smart Indent Generation

- **Purpose:** To manage internal demand with precise item identification and automated replenishment.
- **Auto-Indent Generation (Inventory Thresholds):**
    - **Trigger:** The system continuously monitors "Available Stock." When stock levels fall below the predefined `minStockLevel` in the Item Master, an Auto-Indent is generated.
    - **Multi-Supplier Price Logic:** If a component is associated with multiple suppliers in the **Supplier Rate Card**:

        - **Primary Selection:** The system defaults to the supplier marked as "Preferred" or the one with the lowest "Agreed Price."
        - **Review Queue:** The Auto-Indent is placed in an "Approval Pending" state. The procurement manager can view a comparison table showing all active suppliers for that item, their respective rates, and lead times before finalising the PO.
        - **Quantity Calculation:** The indent quantity is calculated as: `Max Stock Level - Current Available Stock`.
- **Manual Entry (On-Demand):** Authorised users select from the Master list for non-routine or project-specific requirements.
- **Search & Selection Logic:**
    - **Unique Identification:**
        - If Hex-tagged: Search by **Hex Code**.
        - If Bulk/Standard: Search by **Item Code**.
        - If Product: Search by **Serial Number**.
    - **Smart Search:** Search by **Item Name**, **Description**, **Make**, or **Unique ID**.
    - **Interactive Dropdown:** Displays: `[Unique ID] + [Item Name] + [Make] + [Description] | [Available Qty]`.
    - **Example Selection:** `R102505 + 10K, 1/4W-5% + Electra + Resistor 10K, 1/4W-5% | Qty: 48`

B. Auto-PO (Purchase Order) Mapping & Multi-Supplier Splitting

- **Purpose:** To ensure pricing accuracy, reduce manual work, and handle multi-vendor procurements from a single request.
- **PO Generation Logic:** When an indent (Auto or Manual) is approved, the system evaluates the supplier assigned to each individual line item.
- **Supplier Grouping:** If a single indent contains items sourced from multiple vendors (e.g., Resistors from Supplier A and IGBTs from Supplier B), the system automatically splits the indent.
- **Name Translation (Supplier Alias):**
    - When generating the physical PO document, the system checks the **Supplier Rate Card**.
    - If a **Supplier Item Name** is configured for that vendor, the PO will print the supplier's specific name for the item.
    - If no alias is found, it defaults to the internal **Item Name**. This ensures the supplier receives an order using their own terminology while Techser maintains its internal naming consistency.

- **Multi-PO Creation:** The system generates separate PO drafts for each supplier.
    - **Example:** One Indent -> PO #001 (Supplier A) + PO #002 (Supplier B).
- **Price Fetching:** For each generated PO, the system automatically fetches the "Agreed Price" from the **Supplier Rate Card (Agreement)** associated with that specific vendor.
- **Review & Approval:** All generated POs are displayed in a central dashboard for review, allowing the manager to approve each vendor-specific order separately. Rates remain non-editable to prevent unauthorised changes.

C. Supplier Rate Card (Contract Management)

- **Configuration Source:** This data is managed within the **Supplier Configuration** page, where the Admin or Procurement Head registers authorised vendors.

- **Association Logic (Component-to-Supplier):**
    - Association is established via the **Supplier Management UI**.
    - When a new Supplier is configured, the user "maps" items from the **Component Master** to this vendor.
    - This method is preferred for system efficiency as it allows bulk price updates and lead-time adjustments per vendor without cluttering the technical Component Master database.
- **Data Structure:** For every Item Code, the system maintains a list of approved vendors, including:
    - **Internal Item Code:** Linked to the Techser Master DB.
    - **Supplier Item Name:** The alias name the vendor uses for this part (printed on the PO).
    - **HSN/SAC Code:** The specific tax code associated with this item for this vendor.
    - **Currency:** Dealing with currency (e.g., INR).
    - **Agreed Unit Price:** Fixed price for a specific duration/contract.
    - **Lead Time:** Days required for delivery after PO issue.
    - **Preferred Toggle:** A binary switch to mark the primary vendor for Auto-Indent logic.
    - **Last Updated:** Timestamp and user ID of the person who modified the rate.
- **Commercial Price History:**
    - **Automated Logging:** Every time a unit price is updated in the Supplier Configuration or a GRN is processed with a price variation, the system archives the old price in a **Price History Log**.
    - **Analytics:** Users can view a "Price Trend" graph for any Item Code over a 12-month period to analyse market inflation or supplier price hikes.

D. Automated Supplier Rating (Performance Scoring)

- **Purpose:** To objectively evaluate vendors based on actual operational data.
- **Rating Factors:**
    - **Quality Score (Weight 50%):** Calculated as `(Accepted Qty / Total Inward Qty) * 100`. High RTV counts directly lower this score.
    - **Delivery Score (Weight 30%):** Calculated based on `Lead Time Adherence`. Comparing "PO Delivery Date" vs. actual "GRN Date".
    - **Price Stability (Weight 20%):** Frequency and percentage of price hikes recorded in the Price History.
- **Visibility:** Ratings (1 to 5 Stars) are visible in the **Auto-Indent Review Queue**, helping Managers choose reliable vendors over just the cheapest ones.

3. Production & Traceability

A. BOM (Bill of Materials) Configuration

- **Purpose:** To define the structural blueprint of spares and systems for accurate production builds.
- **Spare PCB BOM (Sub-Assembly):** Defines the technical blueprint for manufacturing or repairing a specific PCB (e.g., TDS 146).
    - **Legend/Designator Mapping:** The BOM does not store Hex Tags. Instead, it maps every component to a **Legend** (e.g., C2, C4, R1, U2, T4).
    - **Stock Linkage Logic:** Users configure the BOM by selecting components from the **Component Master**.

    - **Captured Fields:** For each Legend, the system automatically pulls and links the **Item Code**, **Name**, **Value**, **Description**, **Make**, and **Price** (specific to the selected Make).
    - **Automated BOM Costing:** The system automatically aggregates the prices of all individual components to calculate the total **Material BOM Cost**. This cost is dynamic and updates if a component "Make" is changed or if a new price is negotiated in the Supplier Configuration.
- **System Product BOM (Parent Structure):** Defines the full assembly requirement for finished systems.
    - Includes: Multiple Spare PCBs (each with its own sub-BOM), additional discrete components (Capacitors, IGBTs), and structural hardware.
    - **Total Product Costing:** Sums the Material BOM Costs of all sub-assemblies and individual components.
- **BOM Versioning:** Every BOM has a "Revision Number." The system ensures the operator uses the latest approved revision during the build process.

B. Assembly Build Process with Digital Job Card

- **Purpose:** To execute production orders for both sub-assemblies (PCBs) and final products, while ensuring precise personnel, task, and traceability auditing through a Digital Job Card.
- **Build Initiation (Work Order):**
    - A build is triggered manually or via a production plan.
    - The system loads the Master BOM for the selected item (PCB or Product).
    - A **Digital Job Card** is generated for each build, serving as a dynamic **Task List** that tells the operator exactly what to do next.
- **Task Control & Status Management:**
    - For every task/job in the card (e.g., Wiring, Mounting, Testing), the system provides three control options:
        - **Start:** Initiates the task and starts the session timer.
        - **Pause:** Temporarily halts the timer (for breaks or shifting to another unit).
        - **Finished:** Marks the task as 100% complete and moves the unit to the "Next Task" in the sequence.
    - This helps the system identify exactly how many actual hours are expended on which specific product unit and by which operator.
- **Spare PCB Assembly (Sub-Build):**
    - **Mounting & Component Placement:** Operator places bulk and serialised components onto the PCB as per the Legend mapping.
    - **Scanning:** The operator scans the child Hex codes of traceable components (e.g., specific ICs) as defined in the BOM legends.
    - **Validation:** System checks for correct Component + Legend linkage.
    - **Transformation:** Upon finalisation, components are consumed. The operator affixes a **Pre-printed Hex Tag** (from existing plant stock) to the surface of the PCB and scans it into the system to link the assembly data, transforming its status to 'Available Stock'.
- **System Product Assembly (Final Build):**
    - **Mechanical Mounting:** PCBs are mounted inside the product chassis. Structural components (Heatsinks, Transformers) are installed.
    - **Component Integration:** Scanning and installing High-Value parts like IGBTs and Capacitors.
    - **Connections & Wiring:** Inter-component wiring is performed. The system requires confirmation that all BOM-defined connections are completed.
    - **Traceability Scanning:** Scanning the Hex Codes of Spare PCBs and other traceable hardware into the parent serial number record.
- **Testing & QA Phases:**

    - **Standard Testing:** Functional verification of the unit.
    - **Load Testing:** Stress testing the unit under full electrical load.
    - **Personnel Tracking:** The system allows for **different operators** to be recorded for the Build, Testing, and Load Testing phases. Each stage is time-stamped with the specific Operator ID.
    - **Digital Job Card Tracking:** Testing and Load Testing sessions are logged with start/stop times
- **Parallel Job Management**
    - Operators can work on multiple products simultaneously.
    - If logged into multiple Job Cards, overlapping timestamps are recorded for accurate multi-unit labour allocation.
- **Transformation & Stock Update**
    - Upon passing all tests, the system consumes constituent parts.
    - A Master System Serial Number is generated.
    - Final product status is updated to Available Stock.
    - Digital Job Card is closed, serving as a complete audit trail of tasks, personnel, and timestamps.

C. Parent-Child Genealogy Mapping

- **Purpose:** To create a "Digital Birth Certificate" for every finished product based on its BOM and Job Card history.
- **Logic:** During the build process, the system enforces the linkage of child items to the parent item.
    - **Example (PCB):** Linking bulk ICs and serialised IGBTs to the PCB's unique Hex ID.
    - **Example (System):** Linking the completed PCB (Hex ID) and external transformers/capacitors to the Parent System Serial Number.
- **Result:** The parent Serial Number is permanently linked to these child tags. If a component fails years later, the system traces it back through the build record to the specific GRN, Invoice, and Supplier.

D. Labour Cost Calculation (Conversion Charges)

- **Purpose:** To calculate the accurate cost of human labour for units built over several days, often involving multiple operators.
- **Digital Job Card Logic:**
    - The system retrieves all "Sessions" (Start to Pause/Finish) linked to a specific Job Card.
    - For each session, it calculates the duration: (Session End - Session Start).
    - **Parallel Activity Factor:** If an operator is working on 5 units in parallel for 1 hour, the system intelligently allocates 12 minutes to each unit's Job Card to prevent over-costing.
- **Cost Calculation Formula:**
    - Total Labour Cost per Unit = Σ (Session Duration × Operator's Hourly Rate).
    - **Operator Rate:** Fetched from the **Employee Master** based on the specific operator who performed each session.
- **Workforce Analytics:**
    - Management can generate reports showing total hours worked by an individual across different Job Cards to evaluate productivity and factory utilisation.

4. Logistics & Distribution

A. Automated Dispatch Register

- **Purpose:** Real-time logging of all outgoing stock.
- **Logic:** Every time a product is dispatched against a Transfer or Sales Indent, the system automatically logs the date, destination, vehicle details, and the authorising admin.

B. Logistics Surcharge Allocation

- **Purpose:** To ensure transport costs are factored into the product's final value.
- **Logic:** The logistics team enters fuel, toll, and driver expenses for a specific trip. The system divides the total trip cost by the number of units shipped and adds a "Logistics Surcharge" to each product's landed cost.

5. Financial & Costing Engine (The Core Intelligence)

A. Rolling Overhead Costing

- **Problem:** Current month utility bills (Electricity/Rent) arrive after the products are already built and shipped.
- **Solution:** Current production uses the **Average Overhead per Unit** from the **Previous Month** as a baseline.
- **Result:** Management has a realistic "Transfer Price" for daily operations.

B. Monthly True-Up (Variance Analysis)

- **Purpose:** To reconcile estimated costs with actual expenses.
- **Logic:** Once actual bills are entered at month-end, the Admin triggers the "Reconcile" function. The system calculates the gap (Variance): Actual Cost - Estimated Cost.
- **Reporting:** Generates a variance report showing profit/loss impact for every Serial Number produced that month.

C. Category Master & Ledger Control

- **Purpose:** Custom financial categorisation.
- **Logic:** Admin creates expense categories (e.g., "Machine Maintenance," "Office Tea").
- **Costing Toggle:** Each category has a switch: Include in Product Costing. If ON, expenses in this category increase the product Transfer Price.

6. Automation & Security Logic

A. Payment Ageing Analysis Engine

- **Purpose:** Cash flow management.
- **Logic:** Daily cron-job categorises all unpaid invoices into buckets: 0-30, 31-60, 61-90, and 90+ days past the invoice date.
- **Automation:** Sends a tabular summary to the Finance team via **WhatsApp and Email** every morning.

B. Strict Factory Isolation (Multi-Tenancy Middleware)

- **Purpose:** Data security between Bengaluru and Parwanoo units.
- **Logic:** Every database query is automatically scoped with factoryId from the active user session.

- **Enforcement:** A user assigned to Factory 'B' cannot see or search any record belonging to Factory 'P', ensuring complete node isolation.